SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 40% Dave's Taxonomy: Manipulation (Level 2) Question Count: 5

Total Marks: 50

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Assessment Tasks Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

Introduction

A hypervisor is software that creates and manages virtual machines (VMs). There are two types: Type-1 (Bare Metal) and Type-2 (Hosted).

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Installation	Runs directly on hardware	Runs on a host operating system
Performance	High	Moderate
Security	More secure	Less secure
Reliability	Very reliable	Depends on host OS
Examples	VMware ESXi, Microsoft Hyper-V, Xen	VMware Workstation, Oracle VirtualBox

Analysis

- **Performance:** Type-1 provides direct access to hardware, reducing overhead and improving CPU, memory, and storage performance.
- **Security:** Since there is no host operating system, the attack surface is smaller, making it more secure.
- **Manageability:** Type-1 supports centralized VM management, live migration, resource allocation, and high availability.

Conclusion

For a cloud data center hosting mission-critical applications, **Type-1 Hypervisor is the better choice** because it offers higher performance, stronger security, better scalability, and easier management.

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Introduction

Cloud providers host multiple customers on the same physical server. Virtualization ensures each workload remains isolated.

Isolation Mechanisms

1. Hypervisor separates each VM.
2. Virtual CPUs and memory are allocated independently.
3. Virtual networking separates traffic using VLANs.
4. Virtual storage prevents unauthorized access.
5. Security policies control user access.

Major Attack Surfaces

1. Hypervisor Attack

- If the hypervisor is compromised, all VMs are at risk.

2. Virtual Network Attack

- Attackers may intercept or manipulate traffic between VMs.

Security Measures

- Keep hypervisor updated.
- Use firewalls and network segmentation.
- Enable encryption for data.
- Apply Multi-Factor Authentication (MFA).
- Regular security monitoring.

Conclusion

Virtualization provides strong workload isolation, but securing the hypervisor and virtual network is essential for protecting cloud environments.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

Given Data

- CPU Utilization = **92%**
- Memory Utilization = **88%**
- Storage I/O Latency = **35 ms**
- VM Boot Time = **170 seconds**

Analysis

CPU (92%)

- CPU is heavily utilized.
- Applications may respond slowly.

Memory (88%)

- Memory usage is very high.
- System may experience swapping and reduced performance.

Storage Latency (35 ms)

- High disk latency indicates storage bottlenecks.
- Read/write operations become slower.

VM Boot Time (170 s)

- Very slow boot time.
- Indicates storage or resource congestion.

Corrective Actions

- Add more CPU cores.
- Increase RAM.
- Upgrade storage to SSD/NVMe.
- Balance workload across multiple hosts.

- Remove unused VMs.

Conclusion

The host is overloaded. Increasing resources and optimizing storage will improve VM performance and reduce boot time.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

Introduction

An SDDC virtualizes compute, storage, and networking, allowing centralized software-based management.

Improvements

1. Compute Virtualization

- Multiple VMs run on one server.
- Resources are allocated dynamically.

2. Storage Virtualization

- Combines multiple storage devices into a single pool.
- Faster provisioning and better utilization.

3. Network Virtualization

- Creates virtual networks without changing physical hardware.
- Improves flexibility and security.

Advantages over Traditional Data Centers

- Faster deployment
- Better scalability

- Lower operational cost
- Automation reduces manual work
- Improved disaster recovery

Conclusion

SDDC increases agility by automating resource management, reducing costs, and improving flexibility compared to traditional data centers.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Problem

Different cloud providers use different identity management systems, causing authentication failures.

Proposed Solution

1. Identity Federation

Allows users to log in once and access multiple cloud services.

2. Single Sign-On (SSO)

One login for multiple cloud platforms.

3. Standard Protocols

SAML
OAuth 2.0
OpenID Connect
Privacy and Security Measures
Multi-Factor Authentication (MFA)
Role-Based Access Control (RBAC)
Encrypt identity information

Audit logs for monitoring

Least privilege principle

Benefits

Better user experience

Strong security

Easy access across clouds

Reduced password management

Improved compliance

Conclusion

Identity federation with SSO, MFA, RBAC, and secure authentication protocols provides secure and seamless access across multiple cloud providers while protecting user privacy